P11 — Stages of PROJECT spacetimes that were paused on November 12, 2026

Provenance required by Hard Rule 7: what is not finished, why, and how to finish it

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1. What this document is for

On Thursday, November 12, 2026, at 15:20 Pacific Standard Time, the author instructed that all running stages be paused and completed later, and that a new piece of work be started in the folder `Revision_Deflation`. This document records, for the stages that were interrupted, (i) what was running, (ii) what state it was in, (iii) what is known already, and (iv) what has to be done to complete it. Nothing in the stages listed here is to be taken as finished.

The state of the repository at the moment of the pause is the commit `3368a72`. All documents, programs and data of that commit are complete in themselves; the stages below were to *check and refine* them.

2. Paused stage A: the independent verification

The instruction. The task specification asks: "Check and verify all of your previous results/conclusions […] and refine and refactor your solution."

What was running. Four independent verification passes had been started after the commit `3368a72`. Each pass was given to a separate worker that had not seen the work being produced, that could read the repository but not change it, and that was to re-derive or re-compute what it checked by its own means. Each was to deliver a report with three parts: findings (error, gap or minor point, each with its exact place, the evidence and a proposed correction), what was verified and how, and what was not checked.

Pass	Scope
A	The two sets of Dirac matrices, the groups they generate, the chiral projectors, the Dirac equations, the classification of the fermion pairs ("dirac matrices"; manuscripts, Secs. II to IV and Appendix A).
B	The Vorton-Scalar model, the exchange amplitudes, the gap equation, the bound pair, the effective action of Cornwall, Jackiw and Tomboulis, the Higgs discussion (Supplements S1, S2, S4; manuscripts, Secs. V to X).
C	Cosmology and dark matter (Supplement S3; manuscripts, Secs. XI and XII and Appendix D).
D	Audit of the documents: every number against the data files, cross-references, reproduction of the data from a clean copy, the symbol rules of P1, the layout, and the rule that the manuscripts do not mention the tools with which they were prepared.

State. The passes had run for about eleven minutes when the service that runs them was restarted. They were started again and, a few minutes later, stopped at the author's instruction. **No pass delivered a report. No finding was received.** Therefore nothing in the project has been confirmed,

and nothing has been refuted, by this stage. The working files of the passes are in a temporary workspace, not in this repository, and should be considered lost.

To complete. Run the four passes again with the scopes of the table; evaluate every finding; correct every confirmed error in the programs and in the documents; record each correction in `provenance/P9_error_log`; write the result as `provenance/P10_independent_verification` (`.md`, `.tex`, `.pdf`), with the findings, their resolution, and what was not checked; add P10 to the list of documents in `README.md`.

3. Paused stage B: fermions bound to a vortex line (a known imprecision)

The statement that is to be corrected. Section V H of both manuscripts and Sec. 8 of Supplement S1 say that the Dirac equation in the background of a vortex line of the Vorton-Scalar with winding number n has "$|n|$ normalizable solutions of zero energy that move along the line at the speed of light", and attribute this to Jackiw and Rossi. The count $|n|$ is correct for *one four-component* Dirac field. It is not the count for the sixteen-component fields of this project. The statement is recorded as error E43 in `provenance/P9_error_log`, with the status "identified, not yet corrected".

What the analysis gives so far. In the sector of fields that do not depend on the extra coordinates, the one-particle Hamiltonian in a Vorton-Scalar background $\Phi_{VS} = (\phi_1 + i\phi_2)/\sqrt{2}$ is

$$\mathsf{h} = \alpha^j p_j + g_{VS}\,\phi_1\,\beta + g_{VS}\,\phi_2\,(i\beta\chi), \qquad j = 1, 2, 3,$$

with the five anticommuting Hermitian matrices $\alpha^1, \alpha^2, \alpha^3, \beta, i\beta\chi$ of either spacetime. For a straight vortex line along $x3$ with one unit of winding, $\phi_1 + i\phi_2 \propto e^{i\varphi}$ in the plane $(x1, x2)$, the solutions of zero energy that do not depend on $x3$ are a common radial function, which decays away from the line, times a constant spinor u that satisfies the two conditions

$$(i\beta\alpha^1)\,u = -u, \qquad (\alpha^2\beta\chi)\,u = -u.$$

The two matrices are Hermitian, have square one and commute. The prototype `provenance/paused/vortex_prototype_algebra.py` evaluates these conditions with the matrices of `data/gamma1644.txt` and `data/gamma1671.txt`. Its results, the same for both spacetimes, all exact:

Quantity	Result
Dimension of the space of spinors u that satisfy both conditions	4
The matrix $i\alpha^1\alpha^2\alpha^3\chi$ (the product of the five matrices)	commutes with all five, is Hermitian and traceless and has square one: it is $\pm P$, the generator of $U(1)_E$
α^3 on the space of the four solutions	equal to $i\alpha^1\alpha^2\alpha^3\chi$; eigenvalues $+1, +1, -1, -1$
The same for the opposite winding (second condition with $+u$)	dimension 4; $\alpha^3 = -i\alpha^1\alpha^2\alpha^3\chi$

Because the energy of such a solution with the wavenumber k_3 along the line is $\alpha^3 k_3$, two of the four solutions move in the direction $+x3$ and two in the direction $-x3$, at the speed of light; the two directions are distinguished by the sign of P. For a general winding number the count per four-component field is $|n|$ (Jackiw and Rossi; an index theorem), so that the sixteen-component field has $4|n|$ zero modes, $2|n|$ in each direction.

A second prototype, `provenance/paused/vortex_prototype_profile.py`, solves the equation of the profile of the vortex, $f'' + f'/\rho - n^2 f/\rho^2 = \frac{1}{2} f(f^2 - 1)$ with $\rho = m_s \times$(distance from the line), by two methods. For $n = 1$ it gives the slope $f'(0) = 0.4123772$, which is the known value $0.58319/\sqrt{2}$ of

the vortex of the Ginzburg-Pitaevskii equation, and the constant 0.0343 in the energy per unit length $\pi v_{VS}^2[\ln(m_s R) + 0.0343]$ inside a radius R, which corresponds to the known form $\ln(1.464\,R/\xi)$ of that equation.

What is not done. These are prototype results. They are not part of the Rust program, they have no unit tests, and the documents have not been changed.

To complete. (1) Add a module `vortex.rs` and a command `spacetimes vortex` to the Rust program: the algebra above for both spacetimes; the profile for $n = 1, 2$; the radial functions of the zero modes; the count of normalizable solutions in each angular-momentum channel; unit tests; data files; one figure. (2) Correct the two passages, and say in Sec. XIII of the manuscripts that the result is now computed. (3) Enter the new symbols in the registry of P1, extend P8, complete entry E43 of P9. (4) Run the fixed point of the symbol checks and rebuild all documents that changed.

4. Other things that were still to be done

- The final report to the author, with three questions: whether the affiliation printed in the manuscripts (“Independent researcher”, taken from the manuscript of the project NCSS) is the one wanted; whether the reference code CLASS may be built and run on this computer (it was not; the comparison uses the spectra stored in the project NCSS); and a reminder that the journal asks authors to state how a manuscript was prepared, while the manuscripts, as instructed, contain no such statement.
- The code was executed in the cloud workspace, not under macOS: the sandbox on the author’s Mac has no Rust compiler. `provenance/P8_code` gives the commands for macOS; they have not been run there.

5. What is *not* affected by the pause

The work in `Revision_Deflation` is a separate piece of work with hypotheses of its own. It uses the programs and matrices of this project, but it does not change any file outside its folder, and its documents state their own results. The pending correction E43 concerns vortex lines, which the revision treats independently.